Sahaj Bhadja

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Summary

Solutions-oriented engineer with 2+ years translating **customer requirements into data-driven technical concepts**; experienced gathering needs, analyzing operational data, presenting proposals, aligning international stakeholders, and supporting project handoff.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Partnered with stakeholders to gather, validate, and interpret **operational and technical requirements**, then designed scalable data-driven platform concepts supporting 500+ users while reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Collected, organized, and analyzed **10,000+ operational records** to support design validation and feasibility checks, implementing reconciliation, failure isolation, and reliability controls that sustained 99%+ performance across 5,000+ monthly workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Translated stakeholder workflows into **clear product concepts and technical designs** covering React interfaces, Flask services, uploads, processing, and reporting, serving 150+ users and 10,000+ daily requests with stable performance.
- Prepared and delivered compelling technical walkthroughs that aligned stakeholders on solution options, incorporating feedback through reviews and documentation while reducing turnaround 60% across customer-facing operations serving 150+ instructors.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Served as a **technical liaison** with Microsoft and GitHub engineers across regions, gathering requirements, validating designs, and aligning deliverables while improving API response times 30% and pipeline efficiency 68%.
- Collaborated with international engineering teams through Agile sprints, documenting technical designs and automating validation across Windows, Linux, and macOS while maintaining 20+ tests with Jenkins, PyTest, and Mockito.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **customized operational system concepts** across eight coordinated services for authentication, records, billing, permissions, and notifications, securely synchronizing 2M+ customer records through REST APIs and real-time updates.
- Analyzed usage data and customer workflows to identify inefficiencies, reducing resource waste 42% and deployment time 50% while clearly documenting process improvements for cross-functional engineering and operations teams.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Worked directly with customers to gather, validate, and present **end-to-end solution designs** covering booking, pricing, payments, notifications, and operational reporting across 50,000+ records with sub-200-millisecond responses.
- Mapped and documented 12+ integration flows into clear technical proposals for project handoff, implementing secure webhooks and real-time synchronization while reducing manual refund handling 70% through practical automation.

Skills

Solution Design: Requirements gathering, business analysis, concept design, feasibility studies, solution proposals, technical presentations, project handoff
Analytics: Data analysis, Excel, PowerPoint, Microsoft Office, SQL, data validation, operational metrics, Tableau
Systems: Process improvement, automation, system design, scalable architecture, reliability, throughput optimization, workflow mapping
Integration: REST APIs, PostgreSQL, SQL Server, Python, technical specifications, documentation
Collaboration: Customer engagement, cross-functional collaboration, international stakeholders, stakeholder alignment, project lifecycle, Agile, relationship building

Projects

HMIS Live | *TypeScript, REST APIs, Sentry*

- Designed a **customer-facing operational workflow solution** with role-based access, dashboards, appointments, REST integrations, and notifications, supporting 100+ daily workflows while reducing record retrieval time 50% and administration 30%.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Analyzed schemas across 10+ data systems and translated operational requirements into practical structured query solutions, supporting exploration, validation, compatibility checks, and reusable documentation for cross-functional technical teams.

Trend Tracker Live | *React.js, Material UI, CNN*

- Built a **retail-facing recommendation solution** with reusable customer interfaces and data-driven personalization, improving recommendation accuracy from 78% to 92% while serving 3M+ users and retaining 200,000+ customers.